

3.1 Unit 1 INFO1 Practical Problem Solving in the Digital World

Introduction

Courses based on this unit should provide students with the opportunity to develop their knowledge and understanding of the development of ICT systems through **practical experience** in using a range of applications software in a structured way. Students should then be able to apply the skills, knowledge and understanding gained from this practical work to the solution of problems.

Range of Software/Problems

Students are expected to make practical use of a range of applications software, hardware and communication technologies to address several different types of problem. The range of applications software used should cover the processing of text, images, numbers and sound.

Students should address several problems. These problems can be selected to meet the individual needs of learners. Examples of suitable solutions to problems could include several from the following:

- an electronic photo album
- a rolling multimedia presentation for a school open day
- a website for a local nursery
- organising a blog for a local councillor
- producing invoices for a small business
- organising a set of podcasts for a teacher
- an interactive multimedia display for a tourist information centre.

Students should be provided with opportunities to develop the following skills:

- analysing
- designing
- implementing
- testing
- evaluating systems.

Assessment

Assessment of this unit is by examination so candidates can show their understanding of the processes they have completed by their ability to repeat those processes (e.g. identifying suitable software to solve a given problem), or by being questioned about what they have done.

Questions will test candidates' practical experience in identifying methods of solution, and the design, implementation, testing, and evaluation of actual solutions based on a variety of applications software that process different types of data. Candidates will also be asked to identify evidence from Sample Work that they have completed and brought into the examination.

Sample Work

Candidates must bring into the examination a sample of their work which addresses one or two problems. This Sample Work must be submitted with the question paper/answer booklet at the end of the examination.

The Sample Work must address at least one problem and provide evidence as follows.

1. A Problem Identification with a list of requirements for that problem. Interpretation of those requirements as Input, Processing and Output.
2. A test plan and clearly annotated samples of testing evidence that is cross-referenced to the test plan.

Items 1 and 2 could be from the same or two different problem solutions.

Candidates are expected to provide between 10 and 20 pages of Sample Work. It must be succinct, clearly laid out and legible to the examiner. Page numbers must be clearly visible. (The purpose of numbering is to facilitate the candidate and the examiner in finding the required evidence quickly and referencing it accurately.) Page numbering should therefore **be sequential from start to end** and not subdivided into sections. It should be noted that, if the pages are not numbered, examiners will not have time to search through the Sample Work for the appropriate references.

Supervision and authentication of Sample Work

The first page of the Sample Work must be the Candidate Record Form which, in advance of the examination, must be completed and signed by the candidate and the teacher responsible for the supervision of the candidate's work in order to authenticate that the Sample Work is the candidate's own. This is a similar procedure to that concerning authentication of coursework; see Section 6.1.

For further details and examples of Sample Work, see the Teacher Resource Bank.

Topic	Key Concepts	Content and Amplification
1. Health and safety in relation to the use of ICT systems	The need for and application of current health and safety legislation that relates to the use of the ICT systems.	
	Health and safety guidelines cover the design and introduction of new software.	
2. Analysis	Problem identification.	Description of a problem that requires solving, including identifying who needs the solution to the problem (the Client), who will use the solution and what their (the Users') skills are, and any further intended audience.
	Produce a list of the Client's requirements.	Precisely what the solution is to produce and how the solution is to work.
	Interpret lists of requirements as input, processing and output for ICT solutions.	What information in what format is required as output? What data is needed to produce the outputs required? What processing is involved in converting the data inputs to information output?
3. Design of solutions	Select design tools and techniques, and apply them to identified problems. Design of data entry into ICT systems.	Design techniques will depend upon the solution chosen. See the Teacher Resource Bank for examples. Design of data capture forms and other input methods e.g. video and audio capture. Design of robust data entry using validation and verification techniques as: - errors can occur as data is entered into an ICT system - data needs to be accurate - data needs to be valid.
	Design of processes.	Design of each process to be implemented. See the Teacher Resource Bank for further details and examples.
	Design of output of information.	Design of output method and format. Information to be produced in suitable formats and with suitable sequencing. See the Teacher Resource Bank for further details and examples.
	Designs need to be used by other people.	Produce designs that could be implemented by a third party.
4. Selection and use of input devices and input media	Select from input devices and input media available, and justify why items selected were suitable to solve each particular problem.	By knowing the following, candidates will be able to select and justify the most suitable input devices and media: - the broad characteristics of current input methods - appropriate situations for use of each method - the input devices and media used for each method of input - any limitations of current input devices and media.

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5. Selection and use of storage requirements, media and devices	Select from storage media and devices available, and justify why items selected were suitable to solve each particular problem.	By knowing the following, candidates will be able to select and justify the most suitable input devices and media: • data needs to be stored for future use • data can be stored at different physical locations from where it may be used • the broad characteristics of current storage media • appropriate contexts for use of media • the devices needed for reading and/or writing data to the storage media • any limitations of current storage devices and media.
6. Selection and use of output methods, media and devices	Select from output methods, media and devices available, and justify why items selected were suitable to solve each particular problem.	By knowing the following, candidates will be able to select and justify the most suitable methods, media and devices: • the broad characteristics of current output methods • appropriate situations for the use of each method • the output devices and media used for each method of output • any limitations on current output devices and media.
7. Selection and use of appropriate software	Identify software available and justify why the selected software was suitable to solve each particular problem. Systems software and applications software	By knowing the following, candidates will be able to select and justify their choice of software: • an awareness of the concept of software, the need for and function of software • the role and functions of systems software and the role and the functions of applications software • appropriate uses of current applications software.
8. Implementation of ICT-related solutions	Production of working solutions.	Use of the hardware and software and appropriate communications technologies as required to provide working solutions to identified problems.
9. Testing of ICT-related solutions	Construct test plans that include a range of suitable test data, identify expected outcomes and test that the client's requirements have been met.	A test plan that shows the design of representative examples of tests to ensure: • validity of data input • accuracy of output • presentation of output • that the solution meets the requirement of the client • that the solution is usable by the end user and/or intended audience • that it incorporates the data to be used for testing the solution and the expected results of each test.
	Follow test plan in a systematic way.	Testing evidence that is cross-referenced to the test plan. Testing evidence that is clearly annotated to show an understanding of the testing process. Interpret the results of tests and take appropriate action. The results of testing evidence are used in the evaluation of the solution.

Topic	Key Concepts	Content and Amplification
10. Evaluation of ICT-related solutions	Assessment of the effectiveness of the solution in meeting the Client's requirements and showing an understanding of what makes an ICT solution effective.	Candidates should ask themselves the following questions: • does the solution do what it is supposed to do? • does the solution do it in the way it is supposed to do it? • is the solution an effective one? • if it is not, then what is wrong with the solution and what would make the solution an effective one?.